order27 RESULTS

PREREG sha256 7ae5821ba48581b6d5783cb591c1ee5dc8a15332ffaf1d034f7fd625be34ece7 (hashed before numerics).

Transport convention (settled)

Γ_g is defined by G_{gα,gβ} = Σ_k (Γ_g)_{jk} G_k for every automorph, det ±1 alike. The transport Γ_g·Cor_C = Cor_{g⁻¹C} holds for improper g too, because its proof never uses det. The order26 result "Γ_h ≠ I" is correct and not a contradiction: Cor_K is Γ_h-fixed without Γ_h being I.

check	result	verdict
Γ_σ = I (exact)	true	PASS
X1 Γ_h = Γ_σΓ_{−I} = Γ_{−I}Γ_σ (exact)	true (Γ_{−I} ≠ I)	PASS
X2 Γ_{−I}² = Γ_h² = I (exact)	true	PASS
X3 [Γ_h, M] = 0 (exact multisets)	0 mismatches	PASS
X4 [Γ_h, Γ_γ] = 0 (exact)	commute	PASS
N1 Cor_{hK} = Cor_{−K} = CO	2.0e−14	PASS
N2 Γ_h·CO = CO	2.9e−14	PASS
N3 order26 Cor_B reproduced	7.2e−15	PASS (consistency)
RG18 orbit {γ^a h^b K, −K} outside span{CO, Γ_γCO}	r_new ≤ 9.8e−15	**VACUOUS by rule** (predicted)

Character projector ranks (γ,h) = (+,+) 36, (+,−) 24, (−,+) 18, (−,−) 27.

Category A (proved; in v10)

Lemma (parity): Cor_{−C} = Cor_C. Proposition: Γ_h = Γ_{−I} ≠ I, Γ_h² = I, it commutes with M and Γ_γ, and Γ_h·CO = Cor_{hK} = CO. Corollary: the ⟨γ,h⟩-orbit of the corner is {CO, Γ_γCO}, so every group average or projection lies in span{Y_+, Y_−}.

Category B

The numbers above.

Verdicts

Cor_{hK} = Cor_K: Category A. RG18: VACUOUS (this is a theorem now, not only numerics). v10: yes, 134 pages. Ξ: not identified. E1–E3 OPEN.

Disclosure: the predictions used v9 facts (Γ_γ² = I), so they are a derivation; N3 is a consistency check.

order27 NOTES

- With Γ_h fixing CO and ⟨Γ_γ,Γ_h⟩ ≅ (Z/2)², the automorph group gives no new law object; group-based geometric maps on A2_corner are exhausted.
- The (−,−) and (+,−) sectors (ranks 27 and 24) are invisible to the corner. The law defect R(CO) is Γ_h-fixed because R commutes with Γ_h, so it lives in the (±,+) sectors.
- Next geometric step (order28, to preregister): leave the corner orbit. Test non-cone geometric objects, namely the A2_lines part, or corner terms of non-automorph unimodular cones (Farey/continued-fraction cones of the reduced cycle of A between K and γ⁻¹K, e.g. cone((1,0),(7,−1)) is hK, so use the next convergent cones). Non-vacuity: object ∉ span{CO, Γ_γCO}.
- Separate DISCOVER_GRADING_PATH / maps stream: not touched (per steering). MAPS_CANDIDATES untouched. No Resonant objects, no bridge claim.
- nice, OMP_NUM_THREADS=1, 3 GB ulimit, ~130 s. No push or sends. New files only.